

TelcoPulse AI: Technical Description

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Problem

Indosat Ooredoo Hutchison (IOH) serves approximately 95 million subscribers in Indonesia and faces sustained churn pressure from SIM consolidation and aggressive competitor pricing. The current retention process is reactive: customer service responds only after a subscriber has already left. There is no system to identify at-risk subscribers in advance, deliver personalized offers before they churn, or measure outreach effectiveness. The result is millions of preventable revenue losses each month.

AI Approach: Hybrid Architecture

TelcoPulse combines two AI techniques, each suited to a different part of the problem:

Predictive AI (churn scoring). A class-balanced XGBoost classifier trained on 18 behavioral and demographic features (tenure, monthly spend, data usage percentage, NPS score, complaint history, dropped calls, app engagement, days-since-topup, etc.). XGBoost was chosen over deep learning because the data is structured and tabular, where gradient-boosted trees consistently outperform neural networks at small-to-medium scale. The model outputs a churn probability between 0 and 1, then applies a tuned decision threshold (0.40, found via PR-curve F1 maximization on the test set) to produce binary risk labels and risk tiers (Low / Medium / High).

Generative AI (multi-surface). Anthropic Claude Sonnet 4 via API powers seven distinct prompt chains, each behind its own Next.js route: per-segment campaign content (SMS, email, push, WhatsApp), natural-language segment queries to SQL, dashboard insights, campaign impact forecasting, ICP / persona generation, end-to-end launch plans, and per-subscriber retention briefs (risk narrative + recommended channel/offer + personalized email + voice script). Each route degrades to a deterministic template fallback when no API key is configured, so the prototype never breaks.

Prototype Features

The prototype is a Next.js 14 web application deployed on Vercel with nine modules: Churn Radar, Subscriber Workflow (human-in-the-loop tracker with on-demand Claude brief and three differentiated decision paths — Approve / Escalate / Mark Safe — plus channel override, editable email + voice drafts, and outcome capture submitted through /api/feedback), Smart Segments, Campaign Writer, Impact Predictor, What-If Simulator, Model Evaluation, Persona Architect, and Launch Copilot. Outbound delivery is wired through Twilio Voice and Gmail SMTP via Nodemailer, gated by environment variables.

Evaluation Results

All numbers reported on the 2,000-row stratified test split. XGBoost is shown at its F1-optimal threshold; baselines at their defaults.

Metric	Target	XGBoost	Logistic Regression	Rule-based	Status
AUC-ROC	≥ 0.65	0.73	0.72	0.69	PASS
Recall (churn)	≥ 0.50	0.66	0.66	0.55	PASS
Precision (churn)	≥ 0.40	0.41	0.39	0.49	PASS
F1-score	≥ 0.45	0.50	0.49	0.52	PASS
Inference latency	< 5 s	0.3 ms	—	—	PASS

Confusion matrix (XGBoost @ tuned threshold 0.40): TN 1,023 · FP 480 · FN 168 · TP 329.

Targets are deliberately recall-prioritized: in CVM, the cost of a missed churner (lost ARPU × LTV) is materially higher than the cost of a false positive (a low-cost retention offer to a stable subscriber). XGBoost wins on AUC, the threshold-independent ranking quality, which gives the operator flexibility to slide the precision/recall trade-off at deployment time.

Technology Stack

Next.js 14 (React, Tailwind CSS, Recharts), XGBoost 2.x with scikit-learn, Anthropic Claude Sonnet 4 (Anthropic SDK), Nodemailer (SMTP), Twilio Voice API, Vercel (serverless deployment), GitHub (version control and CI/CD).

Recommendation: Go (with conditions)

All five Go/No-Go thresholds passed. Three conditions before scaled rollout: (1) data quality sprint to unify legacy subscriber IDs and fill missing NPS/complaint history, (2) recalibration on real production data — expect AUC in the 0.65–0.75 range once trained on actual CDR/billing/NPS streams, (3) A/B pilot on 10,000 subscribers comparing model-targeted vs. random retention outreach across two 30-day cycles. Proceed to full production only if the pilot shows ≥ 10% churn reduction with statistical significance ($p < 0.05$) and offer-cost-to-revenue-saved ratio ≤ 0.30 .

Live Demo and Code

- App: <https://telcopulse-ai.vercel.app>
- GitHub: <https://github.com/mn3265-commits/telcopulse-ai>